Aaditya Kumar Singh

Data Engineering and Data Science | ETL/ELT Pipelines, Data Modeling, Machine Learning, AI Tools
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Data engineer with six years building the ETL pipelines and data models financial institutions ran on, now in a data science master's at Northeastern. Optimized pipelines processing 500K+ records daily at 25% higher throughput, refactoring that cut query execution time 40%, and automation that removed 60% of a manual reconciliation workload at 99.5% uptime. Clients included CitiBank and Commonwealth Bank of Australia. Available for a Northeastern co-op from January 2027 on CPT.

Education

Maulana Abul Kalam Azad University of Technology | B.Tech Computer Science and Engineering *2016 - 2020*
Coursework: Data Structures and Algorithms; Operating Systems; Computer Networks; Databases

Northeastern University | M.S. Data Science | GPA 4.00/4.00 *Jan 2026 - May 2028*
Coursework: Essentials of Data Science; Algorithms; Database Management Systems

Data Engineering Experience

Senior Data Management Analyst | Synechron, India | Client: Commonwealth Bank of Australia *Mar 2024 - Sep 2025*

- Optimized 12+ data pipelines and ETL processes in SQL, processing 500K+ records daily across twelve source systems that disagreed with each other, profiling each feed before it landed rather than after something broke.
- Refactored 50+ stored procedures across Oracle and Teradata by reading execution plans to find full scans and bad join orders; execution time fell 40%.
- Resolved 30+ production incidents through root cause analysis and Pega PRPC, documenting each in the service desk.
- Converted business stakeholder requirements into technical specifications across 30+ cross-functional reviews.
- Queried large-scale datasets in Amazon S3 through AWS Athena, cutting ad hoc query time 30% for analysts.
- Led a 5-member engineering team as acting lead, directing stand-ups, code reviews and sprint deliverables.

Team Lead / ETL Developer | Empaxis Data Management, India | Client: CitiBank *Dec 2019 - Mar 2024*

- Directed a 6-member team building end-to-end ETL for the CitiBank trade reconciliation platform at 200K+ daily transactions, owning timelines, deliverables and the reconciliation rules themselves.
- Architected extraction and transformation pipelines in Informatica and PL/SQL, cutting the settlement breaks that manual work caused.
- Implemented normalized relational schemas and indexing strategies in Oracle, accelerating reporting queries 35%.
- Mentored 10+ junior engineers through onboarding and authored the standardized technical runbooks the team used.

Selected Data Projects

Streaming Lakehouse: Change Data Capture on Postgres *2026*
PostgreSQL, Debezium, Kafka, Apache Spark, Apache Iceberg, Trino, dbt, Docker · Code: github.com/aaditya2504/streaming-lakehouse

- Architected a low-latency CDC lakehouse across 8 containerized services, isolating OLTP source systems from query load.
- Modeled Silver and Gold dbt layers on Trino across 3 medallion layers with window-function dedup, SHA-256 PII masking and automated quality tests, so every output traces back to the record that produced it.

Restaurant Analytics Database: Schema Design and Concurrency *2026*
MySQL, Aiven Cloud, R, Apache Airflow, Python, R Markdown, SQLite

- Designed a 3NF MySQL schema and built R ETL loading 209K+ visits from CSV and SQLite sources, then ran 4 concurrent loaders with and without transactions to expose the race conditions causing duplicate inserts.

Social Determinants of Mental Health *2026*
Python, pandas, scikit-learn, XGBoost, survey-weighted regression · Code: github.com/aaditya2504/brfss-mental-health-analysis

- Analyzed 200K+ CDC BRFSS respondents across 33 states with survey-weighted regression and ANOVA, engineering 15+ determinants.
- Benchmarked random forest and XGBoost against regression baselines, identified loneliness as the top modifiable factor, and wrote the findings up for readers with no statistics background.

Technical Skills

Languages	Python (pandas, NumPy, scikit-learn, XGBoost), SQL, PL/SQL, R, Java, Bash, Shell Scripting
Data engineering	ETL/ELT, Apache Spark (PySpark), Kafka, Debezium, Airflow, dbt, Informatica, CDC, data lineage
Databases and modeling	Oracle, Teradata, PostgreSQL, MySQL, Snowflake, Redshift, MongoDB, star schema
Cloud and DevOps	AWS (S3, EC2, RDS, Lambda, Redshift, Glue, Athena), Azure, Docker, Git/GitHub, CI/CD, Linux, Jira
Analytics and BI	Tableau, Power BI, Excel, dashboards and reporting, exploratory data analysis, data quality
AI and ML	Claude and OpenAI APIs, prompt engineering, structured output, LLM automation, scikit-learn
Certifications	AWS Certified Data Engineer Associate (May 2025), SQL HackerRank

Additional Experience

Teaching Assistant | Northeastern University, Khoury College | Essentials of Data Science *Sep 2026 - Present*

- Answer questions from 20+ students on Python and statistics, and grade 2 assignment cycles per week.